

1. Summary of Approach

This report presents the findings of a study investigating whether portfolio concentration measures can predict elevated risk of a concentration-amplified loss, developed to support STADIOEquities' goal of identifying clients whose portfolios are dangerously concentrated before losses occur. Since STADIOEquities' own client data was not yet available, this study validated its approach using SEC Form 13F institutional holdings data as a publicly available proxy for client portfolio behaviour.

Five portfolio-level features were engineered for each institution-quarter: the Herfindahl-Hirschman Index (HHI), top-holding percentage, number of distinct holdings, sector HHI, and total portfolio value. A binary risk label was constructed by computing each portfolio's concentration-weighted excess return relative to the S&P 500 benchmark over the following quarter, using a 45-day filing-date lag to reflect the earliest point this information could genuinely have been acted upon, and excluding implausible single-quarter returns identified as data artifacts. Portfolios whose weighted excess return fell below -5% were flagged as high-risk, a threshold selected empirically after testing several candidates against the actual data distribution. Two models were trained, tuned via cross-validated grid search, and compared: Logistic Regression (an interpretable baseline) and XGBoost (a gradient-boosted ensemble).

2. Recommendation

Both models achieved strong, broadly comparable ranking performance on the held-out test set, with ROC-AUC scores of 0.921 (Logistic Regression) and 0.926 (XGBoost). The two models diverge meaningfully on recall: XGBoost correctly identifies 107 of 114 genuinely at-risk portfolios (recall 0.94), compared with 89 of 114 for Logistic Regression (recall 0.78), at the cost of a higher false-positive rate (1,044 vs. 586).

XGBoost is recommended as the preferred model. Beyond its higher recall, directly relevant given the cost of missing a genuinely at-risk client, XGBoost also produces a more robust account of which factors drive risk. Logistic Regression required carefully

tuned regularisation to overcome a 0.96 correlation between HHI and top-holding percentage, which otherwise produced an unstable, counter-intuitive coefficient sign for HHI; even after tuning, this necessary regularisation came at a direct recall cost. XGBoost, as a tree-based ensemble, handles this same correlation without requiring an equivalent trade-off: HHI and top-holding percentage jointly account for approximately 77% of feature importance, confirming portfolio concentration as the dominant predictive signal without compromising the model's ability to catch true positives.

3. How the Model Could Be Improved

Several further improvements remain genuinely open, beyond what has already been addressed through hyperparameter tuning and cross-validation. First, this study's price data exhibited a documented survivorship bias: securities that were delisted or failed outright, disproportionately likely to represent genuine loss events, were more likely to be excluded due to missing historical price data. Two independent data sources were tested to address this; neither provided reliable, complete coverage for delisted securities, confirming this as a genuine limitation of freely available data rather than an oversight. Sourcing historical price data from a provider with dedicated delisted-security coverage would likely improve detection of the most severe risk cases the model is currently most likely to miss.

Second, the classification threshold could be tuned to STADIOEquities' actual review capacity, rather than the models' default 0.5 probability cutoff, allowing a deliberate, informed trade-off between catching more true positives and generating fewer false alarms. Formal statistical testing supports this picture: McNemar's test confirms the two models' classification patterns differ significantly ($p < 0.0001$), while DeLong's test finds no significant difference in their underlying ROC-AUC ($p = 0.210$), together indicating the models rank risk comparably well overall, but differ meaningfully in which specific cases they classify correctly at a fixed threshold, a difference favouring XGBoost's recall on the costly positive class. Finally, the label is computed relative to a single, flat S&P 500 benchmark; a sector-adjusted benchmark would more precisely isolate concentration-specific risk from general sector-wide market movement.

4. Adapting the Model to STADIOEquities' Own Data

This study validated its methodology using SEC Form 13F institutional holdings data as a proxy, since STADIOEquities' own client data was not yet available. Applying this model to STADIOEquities' retail client base, once the data described in the data request is received, would require several adaptations. Institutional portfolios are substantially larger, in both position count and value, than a typical retail account; feature thresholds and scaling would need recalibrating against STADIOEquities' actual client-level distributions.

More significantly, this study's label is a proxy, a portfolio's concentration-weighted underperformance relative to a benchmark, since no real-world outcome data was available. STADIOEquities' own data request explicitly includes support and complaint history, which would allow the model to instead be trained or validated against genuine client outcomes, such as whether a client's complaint referenced an unsuitable, concentrated position, or whether a highly concentrated client's account subsequently went dormant, meaningfully strengthening the model's real-world validity beyond what a market-return proxy alone can demonstrate. Finally, appropriate anonymisation and POPIA-compliant handling, as noted in this project's RAAIDD log, would be a prerequisite for any such implementation.

5. Alignment with the Literature

This study's findings are strongly consistent with the literature reviewed in Part A. Kacperczyk, Sialm and Zheng (2005) validate the use of a Herfindahl-style concentration index as a defensible, standard measure of portfolio concentration, directly supporting this study's use of HHI as a core feature; their finding that concentration is associated with better average performance also clarifies why this study deliberately targeted the downside tail of outcomes, rather than average returns. Yang et al. (2021) similarly used a modified Herfindahl index as a diversification-evaluation metric paired with drawdown, a pairing methodologically close to this study's own concentration-plus-loss label design. Their approach differs meaningfully in method, however: Yang et al. use portfolio optimisation to actively construct diversified portfolios, whereas this study applies

supervised classification to predict risk in portfolios as already held, a fundamentally different task suited to STADIOEquities' need to flag existing client behaviour rather than prescribe new allocations. Chen et al. (2023), whose study applied machine learning directly to SEC 13F data, provides the closest direct precedent: their use of a forward-quarter return as the prediction target, specifically to avoid look-ahead bias given the delay between a reporting quarter and its actual filing date, is precisely the reasoning this study applied in incorporating a 45-day filing-date lag into its own label construction. Two differences remain worth noting: Chen et al. predict a continuous return and construct quantile portfolios, whereas this study frames the problem as binary classification, better suited to a flag-or-no-flag intervention decision; and their dataset includes an actual filing-date field, while this study's dataset does not, meaning the 45-day lag applied here is a defensible approximation of that same principle, not an identical implementation.

Both models' results reinforce this alignment: HHI is a positive, correctly-signed predictor of risk in both the tuned Logistic Regression model and the XGBoost model, where HHI and top-holding percentage jointly dominate feature importance. This supports the central premise shared across the reviewed literature: portfolio concentration is a genuine, measurable, and predictively useful signal of risk, and machine learning approaches applied to institutional holdings data, including 13F-derived data specifically, can meaningfully extract this signal.

References

- Chen, Y., Song, Y., Qian, S. and Fleiss, A. (2023) Applying Machine Learning to SEC 13F Investment Manager Filings for Portfolio Construction and Rebalancing. SSRN Working Paper.
- Kacperczyk, M., Sialm, C. and Zheng, L. (2005) 'On the Industry Concentration of Actively Managed Equity Mutual Funds', *The Journal of Finance*, 60(4), pp. 1983–2011.
- Yang, Y., Zhao, L., Chen, L., Wang, C. and Han, J. (2021) Portfolio Optimization with Idiosyncratic and Systemic Risks for Financial Networks. arXiv preprint.